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EXP7:

```
#include <stdio.h>
#include <string.h>

/* cipher symbols may be multi-byte UTF-8 (e.g. ‡ † ¶ —), so we split into tokens */
#define MAXS 600

const char *sym[] =
{"5","3","‡","†","0",")","6","*",";","4","8","2",".","¶","]",":","(","9","?","—","1"};
const char  plain[] = {'a','g','o','d','l','s','i','n','t','h','e','b','p','v','w','y','r','m','u','c','f'};
#define NSYM 21

int main() {
    char ct[2000];
    char *tok[MAXS];
    int  len[MAXS], n = 0, i, j, cnt[NSYM] = {0};
    char *p;

    printf("Enter ciphertext: ");
    scanf("%1999s", ct);

    /* split into UTF-8 characters */
    p = ct;
    while (*p) {
        int l = 1;
        unsigned char c = (unsigned char)*p;
        if (c >= 0xF0) l = 4; else if (c >= 0xE0) l = 3; else if (c >= 0xC0) l = 2;
        tok[n] = p; len[n] = l; n++;
        p += l;
    }
```

```
    /* step 1: frequency analysis */
    for (i = 0; i < n; i++)
        for (j = 0; j < NSYM; j++)
            if ((int)strlen(sym[j]) == len[i] && strncmp(tok[i], sym[j], len[i]) == 0) cnt[j]++;
    printf("\nSymbol frequencies (most frequent symbol is probably 'e'):\n");
    for (i = 0; i < 4; i++) {
        int best = 0;
        for (j = 1; j < NSYM; j++) if (cnt[j] > cnt[best]) best = j;
        printf("  %s  occurs %d times\n", sym[best], cnt[best]);
        cnt[best] = -cnt[best];
    }

    /* step 2/3: apply the deduced key ( 8=e, ;=t, 4=h, ... ) */
    printf("\nDecrypted message:\n");
    for (i = 0; i < n; i++) {
        char out = '?';
        for (j = 0; j < NSYM; j++)
            if ((int)strlen(sym[j]) == len[i] && strncmp(tok[i], sym[j], len[i]) == 0) out = plain[j];
        putchar(out);
    }
    printf("\n");
    return 0;
}
```

OUTPUT:

```
Enter ciphertext: 53‡‡†305))6*;4826)4‡.)4‡);806*;48†8¶60))85;;]8*;:‡*8†83(88)5*†;46(;88*96*?;8)
*‡(;485);5*†2:*‡(;4956*2(5*—4)8¶8*;4069285);)6†8)4‡‡;1(‡9;48081;8:8‡1;48†85;4)485†528806*81(‡9;
48;(88;4(‡?34;48)4‡;161;:188;‡?;

Symbol frequencies (most frequent symbol is probably 'e'):
  8  occurs 34 times
  ;  occurs 27 times
  4  occurs 19 times
  )  occurs 16 times

Decrypted message:
agoodglassinthebishopshostelinthedevilsseattwentyonedegreesandthirteenminutesnortheastandbynort
hmainbranchseventhlimbeastsideshootfromthelefteyeofthedeathsheadabeelinefromthetreethroughthesh
otfiftyfeetout
```